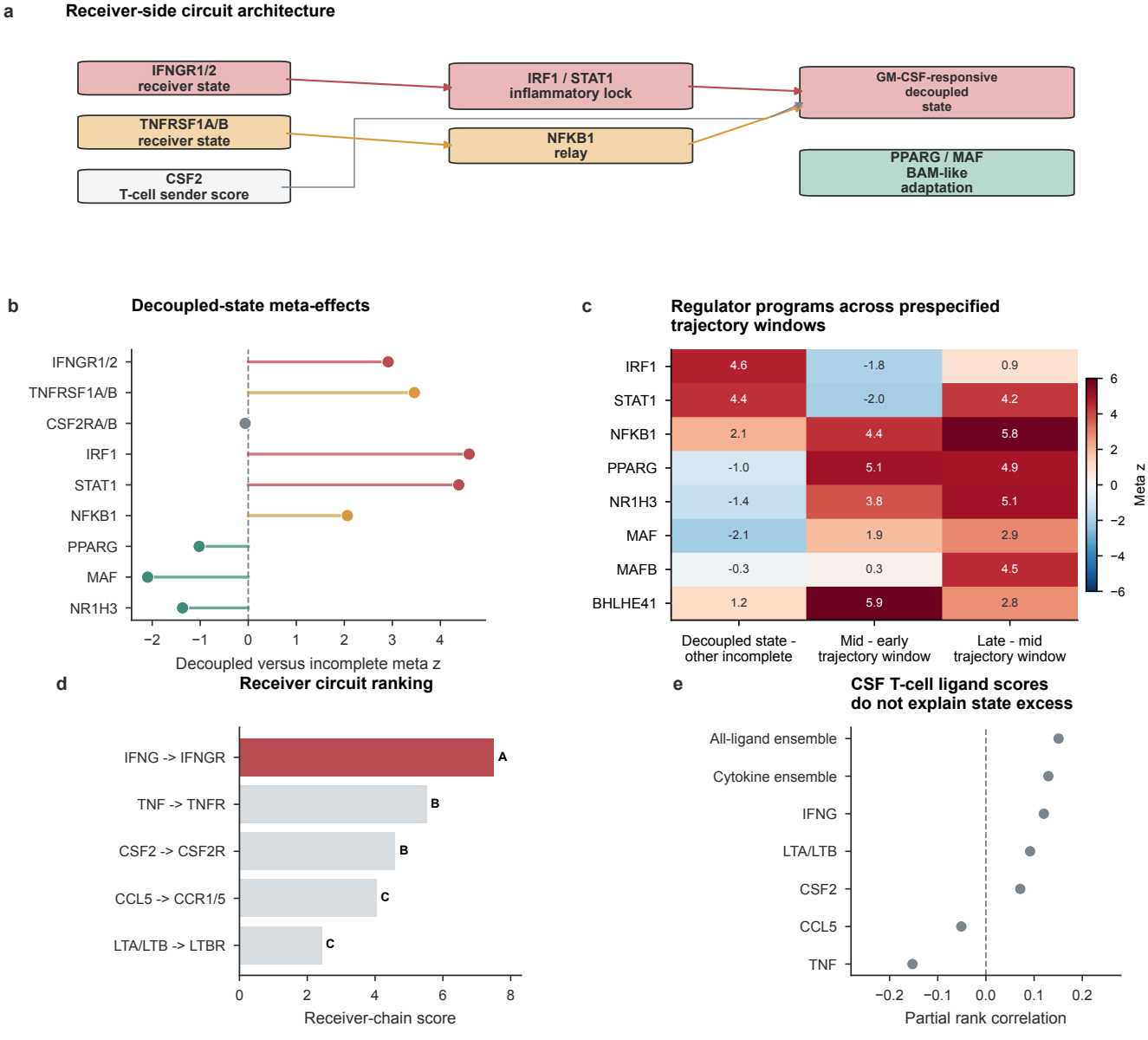

a
Receiver-side circuit architecture
IFNGR1/2 receiver state
TNFRSF1A/B receiver state
CSF2 T-cell sender score
IRF1 / STAT1 inflammatory lock
NFKB1 relay
GM-CSF-responsive decoupled state
PPARG / MAF BAM-like adaptation
b
Decoupled-state meta-effects
IFNGR1/2
TNFRSF1A/B
CSF2RA/B
IRF1
STAT1
NFKB1
PPARG
MAF
NR1H3
−2 −1 0 1 2 3 4
Decoupled versus incomplete meta z
c
Regulator programs across prespecified trajectory windows
IRF1 4.6 -1.8 0.9
STAT1 4.4 -2.0 4.2
NFKB1 2.1 4.4 5.8
PPARG -1.0 5.1 4.9
NR1H3 -1.4 3.8 5.1
MAF -2.1 1.9 2.9
MAFB -0.3 0.3 4.5
BHLHE41 1.2 5.9 2.8
Decoupled state - other incomplete
Mid - early trajectory window
Late - mid trajectory window
Meta z
6 4 2 0 −2 −4 −6
d
Receiver circuit ranking
IFNG -> IFNGR A
TNF -> TNFR B
CSF2 -> CSF2R B
CCL5 -> CCR1/5 C
LTA/LTB -> LTBR C
0 2 4 6 8
Receiver-chain score
e
CSF T-cell ligand scores do not explain state excess
All-ligand ensemble
Cytokine ensemble
IFNG
LTA/LTB
CSF2
CCL5
TNF
−0.2 −0.1 0.0 0.1 0.2
Partial rank correlation